Neel Gali

Data Scientist, Generative AI

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | AI-Powered Customer Intelligence Platform

- Architected and deployed a **LangGraph**-based agentic AI system for complex customer servicing workflows, integrating **Hugging Face** models on **AWS SageMaker** to reduce resolution time by 30%.
- Developed production-grade **Generative AI** solutions using **PyTorch** and **LLMs** on **AWS Lambda** for extracting key information from unstructured documents, improving data extraction accuracy to 95%.
- Implemented comprehensive **AI observability** and monitoring controls for dynamic **LLM**-powered applications using **Weights and Biases Weave**, ensuring system audit trails tracked 100% of autonomous decisions.
- Engineered scalable **VLMs** on **AWS S3** and **EC2** for multi-modal data analysis (text/image) in regulatory compliance, processing 50M records monthly with 98% compliance adherence.
- Curated high-quality, consistent datasets for **LLM** training and evaluation by guiding annotators using internal tools, enhancing model performance and reducing annotation error rates by 20%.
- Evaluated **GenAI** application architectures in production, implementing best practices for **AWS** deployments and ensuring non-deterministic outputs remained within defined risk appetite, preventing 5% potential compliance.
- Developed custom **LlamaIndex** retrieval augmented generation (RAG) applications on **AWS DynamoDB** to enhance context awareness for **LLM** responses, boosting information retrieval accuracy by 25%.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | Enterprise Data Science Platform

- Built and deployed **NLP** models using **Python** and **scikit-learn** on **AWS SageMaker** for sentiment analysis of customer feedback, increasing insights for product teams by 20%.
- Designed and implemented robust **data pipelines** with **AWS Glue** and **Python** scripts to process 100GB daily of unstructured text data, improving data availability for ML models by 40%.
- Developed **Deep Learning** models with **PyTorch** for image classification tasks on **AWS S3** stored data, achieving 92% accuracy in identifying critical components for inventory management.
- Engineered a scalable **REST API** service in **Python** using **FastAPI** on **AWS EC2** for real-time inference of **Machine Learning** models, supporting 5000 requests per minute.
- Optimized **SQL** queries and schema designs for a **PostgreSQL** database, reducing data retrieval times for **big data** analytics from 30 seconds to 5 seconds across complex reports.
- Collaborated on developing **MLOps** practices for model versioning and deployment automation using **Git** and **Jenkins**, decreasing model update cycles from weeks to days by 75%.
- Implemented customer segmentation using **clustering** algorithms in **Python** on **AWS Redshift** data, identifying 5 key customer groups and informing targeted marketing strategies.
- Leveraged **OpenAI API** for text summarization prototypes on internal documents, demonstrating potential to save 10 hours weekly in manual review processes for legal teams.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Enterprise Software Development

- Developed and maintained **Python**-based backend services for an enterprise SaaS platform, integrating with **PostgreSQL** databases to support 10,000 active users with 99.9% uptime.
- Implemented secure **RESTful APIs** using **Flask** on **AWS EC2** for critical data exchange between microservices, improving data consistency and reliability across the platform by 15%.
- Optimized data processing workflows using **Pandas** for large datasets, reducing report generation time from several hours to 30 minutes, enhancing operational efficiency by 80%.
- Managed and optimized **SQL** database performance for a high-traffic application, leading to a 25% reduction in query latency and improved user experience for key features.
- Configured and deployed applications on **AWS S3** and **Lambda** for serverless operations, achieving 99.5% service availability and reducing infrastructure costs by 10% annually.

Skills

Languages: Python, SQL
Frameworks & Tools: PyTorch, Hugging Face, LangGraph, LlamaIndex, Weights and Biases Weave, scikit-learn, Pandas, FastAPI, Flask, Git, Jenkins
Cloud & Infra: AWS, S3, EC2, Lambda, Glue, Redshift, SageMaker, Kubernetes, DynamoDB, PostgreSQL
Concepts: REST APIs, Microservices, CI/CD, Data Pipelines, MLOps, System Design, AI Observability, Regulatory Compliance, Data Annotation, Big Data
AI Tools: LLMs, VLMs, Generative AI, NLP, Deep Learning, Machine Learning, Sentiment Analysis, Classification, Clustering, OpenAI API

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India